

Brief A — Technical: PdM Approach for NBI-P03

Audience: Maintenance/Reliability Engineer | Author: David Kimathi

What we're detecting

NBI-P03 (Nairobi depot) shows a confirmed, statistically significant throughput drop of ~393 barrels/shift versus the depot's other three pumps (Welch's t-test, $p < 0.001$; Cohen's $d = 3.8$ — a large effect). The prototype's goal is to flag at-risk shifts before they're logged as missed_maintenance, using rolling features computed from throughput, temperature, and voltage_count as sensor proxies (no vibration sensor is present in this dataset).

Sensor logic — features engineered

Feature	Window	What it captures
Rolling mean throughput	7 shifts	Smoothed output trend, filters shift-to-shift noise
Rolling RMS of voltage_count	7 shifts	Energy-style summary of SCADA voltage stability
Rolling max-min range (voltage)	7 shifts	Instability/spread, not just level
Rolling std dev of throughput	7 shifts	Output volatility — degrading equipment often gets noisier first

Model & performance

A Logistic Regression classifier (class_weight="balanced") was trained on the four features above to predict a binary Healthy/Failing target, on a 70/30 train-test split (n=170 test shifts).

Metric	Value	Meaning
Sensitivity (recall)	0.899	Share of true failing shifts correctly flagged
Specificity	0.545	Share of true healthy shifts correctly cleared
False Positive Rate	0.455	Share of healthy shifts wrongly flagged
Accuracy	0.876	Below the 0.935 majority-class baseline — see note below

Known limitations — read before deploying

- **Accuracy is misleading here:** because 93.5% of NBI-P03's shifts are already "Failing," a model that always guesses "Failing" would score 93.5% accuracy without doing anything. We deliberately traded accuracy for sensitivity/specificity balance via class weighting — report those two, not accuracy, to stakeholders.
- **Small Healthy sample:** only 37 Healthy shifts exist in NBI-P03's full history, so the specificity estimate (0.545) is provisional, not production-grade.
- **Not yet a true leading indicator:** voltage-based features show no clean pre-break trend and contribute less to the model than throughput itself — meaning the model currently confirms the problem more than it predicts it ahead of time. A real vibration or pressure sensor feed is the recommended next step.

Recommended next step

Pilot this model as an advisory flag (not an automated shutdown trigger) alongside a real vibration sensor retrofit on NBI-P03, so the false-positive rate and true leading-indicator behavior can be validated against ground truth before any automated action is taken.

Brief B — Executive: Value of the PdM Approach

Audience: CFO | Author: David Kimathi

The problem, in business terms

One pump at our Nairobi site has been running in a degraded state since February — producing about 40% less output per shift than it should, on more than 500 shifts so far. Using the dataset's own throughput figures and an illustrative price of \$70/barrel* the lost output this year is worth roughly \$15.8 million — money we are effectively leaving on the table by not catching the problem sooner.

**Illustrative assumption only — this dataset does not include real pricing; substitute your actual realized barrel price for an accurate figure.*

What we built

A predictive tool that watches a pump's daily performance and estimates, before a shift happens, whether that pump is likely to be running in a degraded state — the same kind of early-warning system used across manufacturing and energy operations to catch equipment problems before they become expensive.

What it's worth

- Catches roughly 9 out of 10 truly bad shifts before they happen (in testing) — turning a reactive, after-the-fact discovery into a proactive alert.
- Every bad shift caught early is a chance to schedule maintenance on our terms, rather than losing output first and finding out later.
- Risk reduction: earlier detection limits how long a failing pump runs undetected, protecting both output and the equipment itself from further wear.

The honest caveat

This is a working prototype, not a finished production system. It currently confirms problems slightly earlier than we'd otherwise notice them, but it doesn't yet predict failures several days in advance the way a mature system would. The recommended next step is a small hardware investment — real vibration sensors on the pump in question — to unlock true early warning, which we estimate as a modest cost relative to the value already identified above.

Bottom line

Even this early-stage tool, built on data we already collect, points to a multi-million-dollar-scale opportunity at a single site. The ask is not a large new system — it's approval to pilot this approach on one pump and measure the real-world savings before any wider rollout.

Reflection: Switching Between Technical and Executive Contexts

I ended up writing both briefs roughly together rather than one after the other, which turned out to make the differences between them more obvious in the moment rather than less — I could feel myself wanting to reach for the same sentence in both places and having to stop and rebuild it twice. The hardest part wasn't cutting jargon or picking which numbers mattered; it was getting the tone right. Brief A needed to read with some urgency — this is a real, statistically confirmed problem, don't wait — while Brief B needed to land as reassuring and confident even while saying almost the same thing, since a CFO reading "we found a \$15.8M problem" needs to immediately also hear "and here's exactly what we're doing about it," or the message reads as alarming rather than actionable. Striking that balance — honest about the caveats without undercutting the ask, urgent without sounding panicked — was harder than any of the technical translation work.